

Swapnil Lahiri | AI/ML & Full-Stack Developer

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SUMMARY

AI/ML-focused Computer Science student at KIIT (CGPA: 8.73) with hands-on experience building production-deployed agentic LLM systems, computer vision pipelines, and full-stack applications. Experienced designing multi-node LangGraph pipelines, implementing ML algorithms from scratch, and shipping end-to-end products. Proficient in Python, React, FastAPI, PostgreSQL, and cloud deployment.

PROJECTS

Apr – May 2026

MedPath AI

FastAPI · LangGraph · Gemini 2.5 Flash · Supabase · React 19 · Railway · Vercel

Built for TenzorX / Poonawalla Fincorp Hackathon • Shortlisted among 20,000+ students.

- Built a production-deployed AI health assistant running patient symptoms through a **multi-node LangGraph pipeline** powered by Gemini 2.5 Flash, returning ranked hospital recommendations with real-time cost breakdowns in **<2 seconds**
- Implemented **comorbidity-adjusted pricing** and AI-estimated costs for unmapped procedures with **80% accuracy**
- Integrated a complete **medical loan pre-qualification engine** (FOIR, CIBIL, EMI calculation) with a live PFL officer dashboard
- Secured the platform with Supabase RLS, Storage Bucket Policies, and JWT authentication

Mar 2026

Argus — AI-Powered Online Debate Arena

React · Node.js/Express · FastAPI · HuggingFace · Socket.io

6th Semester Minor Project

- Built a real-time multiplayer debate platform with **server-authoritative state sync** via Socket.io with **<4 second latency** across concurrent live sessions
- AI argument scoring across 4 dimensions using HuggingFace models; **real-time fallacy detection** (ad hominem, straw man, false dichotomy)
- Designed the INTERJECT system — a chargeable ultimate ability that rewards strong argument chains
- Includes live ELO ranking and leaderboard system; spectator mode for passive viewers

Apr 2026

KnowBot — Agentic RAG Research Q&A Chatbot

Python · LangGraph · LangChain · Groq API (LLaMA 3.1) · ChromaDB · Streamlit

Built for LTMindtree Agentic AI Course

- Agentic RAG system for research PDFs using semantic search with citation-backed answers and **self-evaluation** to reduce hallucinations
- LangGraph pipeline with dynamic routing, persistent memory, and faithfulness scoring
- Deployed as an interactive Streamlit app with persistent chat memory and real-time document querying

Feb 2026

Hough Line Detection from Scratch

Python · NumPy · Matplotlib

- Implemented full Canny Edge Detection pipeline manually — no pre-built CV modules
- Built Hough Line Transform algorithm from scratch for lane detection; analysed Hough parameter space to identify dominant lane lines in road imagery

Oct 2025

Rune Tales — Turn-Based RPG

Unity · C# · Murf AI API

- Developed a turn-based roguelike RPG using Unity with full game loop, combat system, and progression
- Integrated AI-generated voiceover narration via the Murf AI API

EXPERIENCE

May – Jun 2026
Remote

AICTE – Cisco Virtual Internship

Industry-recognised program: Networking · Python · AI · Data Science

- Completed 9 courses on Cisco NetAcad; certified in **CCNA** (Intro to Networks, Switching, Routing, Enterprise Networking, Security)
- Certified in Python Essentials 1 & 2, Introduction to Modern AI, Data Science, and Data Analytics
- Built and troubleshoot network topologies using Cisco Packet Tracer — TCP/IP, subnetting, VLANs, routing, switching

Jun 2025
Virtual

Deloitte Australia – Data Analytics Job Simulation

Forage · Forensic technology and data analysis

- Built a data dashboard in Tableau and used Excel to classify data and draw business conclusions
- Completed a simulated forensic data investigation applying real-world analytical techniques

EDUCATION

2023 – 2027
Bhubaneswar

B.Tech, Computational Science — Kalinga Institute of Technology (KIIT)
Completed 6th Semester • **CGPA: 8.73**

2023
Kolkata

High School Diploma — South Point High School
CBSE Secondary: 93.8% • CBSE Higher Secondary: 83%

SKILLS

Programming	Python, Java, JavaScript, C, C#, Lua, SQL
ML & AI	Scikit-learn, TensorFlow, PyTorch, LangChain, LangGraph, HuggingFace Model Evaluation
Computer Vision	OpenCV, YOLO, Canny (from scratch), Hough Transform (from scratch)
Web & APIs	React, Next.js, Node.js/Express, FastAPI, Flask, REST APIs, JWT Auth, Railway, Vercel
Databases	PostgreSQL, MongoDB, Supabase, ChromaDB
Dev Tools	Git, VSCode, Jupyter, Google Colab, Excel, Tableau, Cisco Packet Tracer

CERTIFICATIONS

Udemy: Complete Data Science, ML, DL & NLP Bootcamp (Krish Naik, Nov 2025) • Machine Learning A-Z (Eremenko et al., Aug 2025) • Complete Web Development Bootcamp (Angela Yu, 2024)

Coursera: Business for Good: Fundamentals of Corporate Responsibility • Corporate Governance • Ethical Decision Making for Success in Tech Industry (2026)